BIT 4107 – Mobile Application Development

ENHANCED PROFESSIONAL STUDENT LOGBOOK

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School/Faculty: Computing& Informatics

Programme: Information Technology

Academic Year: 2026

Semester: Two

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Instructions

This logbook is used to document weekly learning activities, practical exercises, skills acquired, challenges encountered, solutions applied, and reflections. Students should complete all sections every week.

Each week to follow remarks in week 1 section.

Autogenerated Table of content

Week 1: Introduction to Mobile Application Development

Practical Activities Undertaken

1. Installed Android Studio.
2. Installed Flutter SDK and Dart SDK.
3. Configured Android Studio with Flutter and Dart plugins.
4. Set up Android Emulator.
5. Created the StarShare project.
6. Created a GitHub repository for project version control.
7. Verified Flutter installation and environment configuration.

8. figure1: Installation and setting up of android studio.

Figure2:Configured Android SDK and Emulator and Verified development environment.

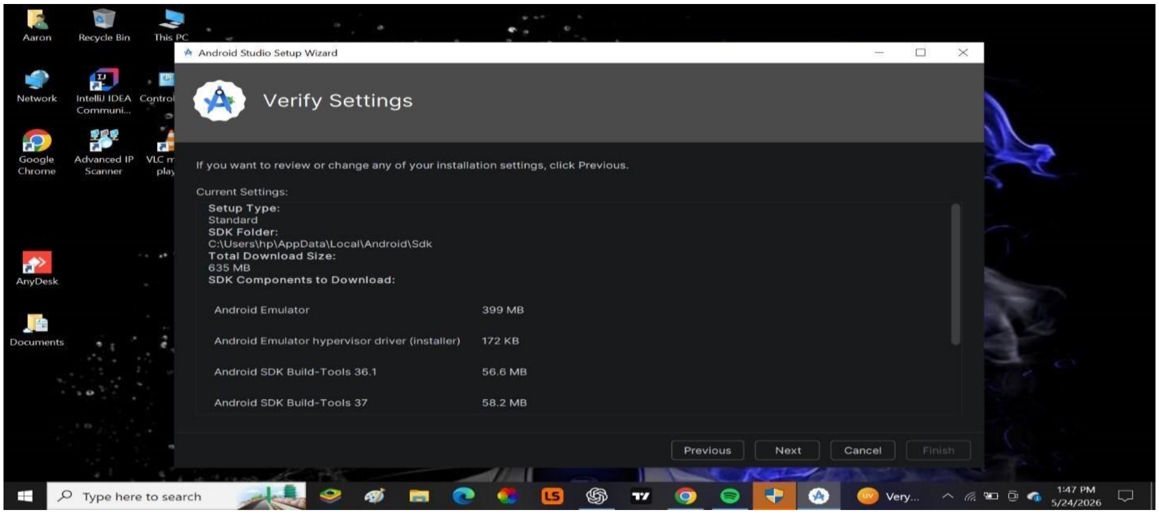
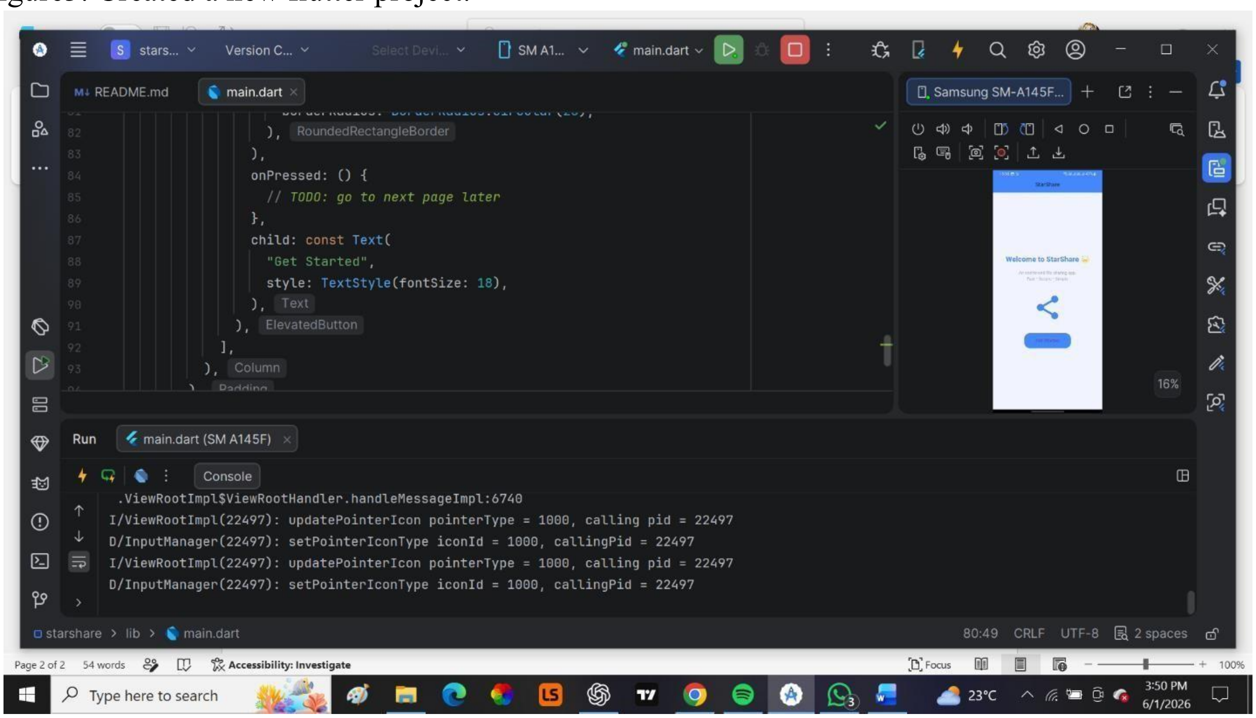
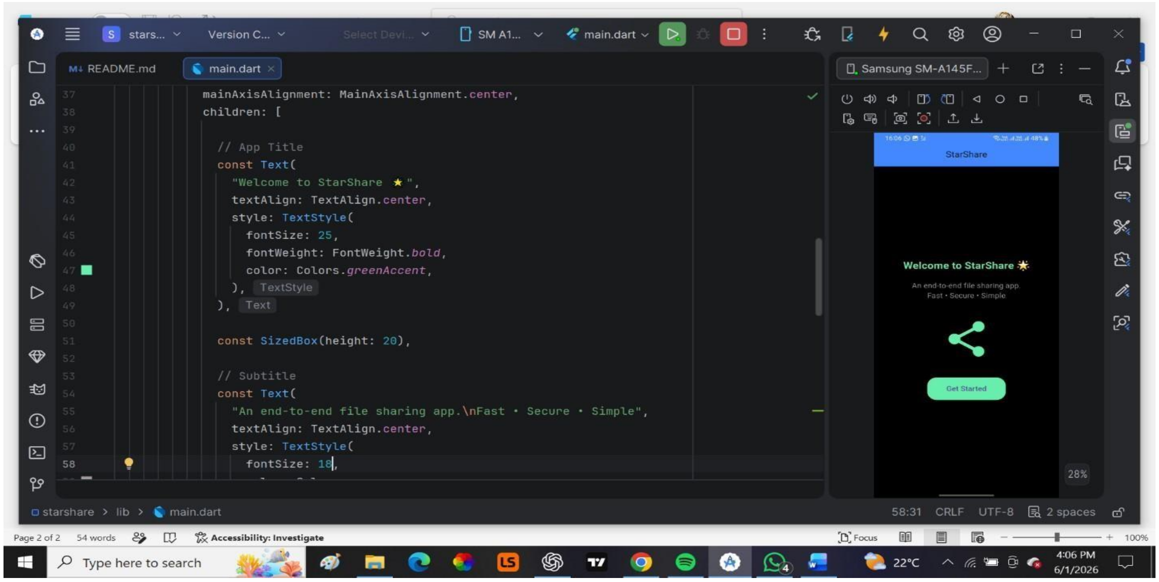


Figure3: Created a new flutter project.



- Figure4: Modified the application title, changed the background colour and adjusted the font size of the application text.



Tools/Software Used

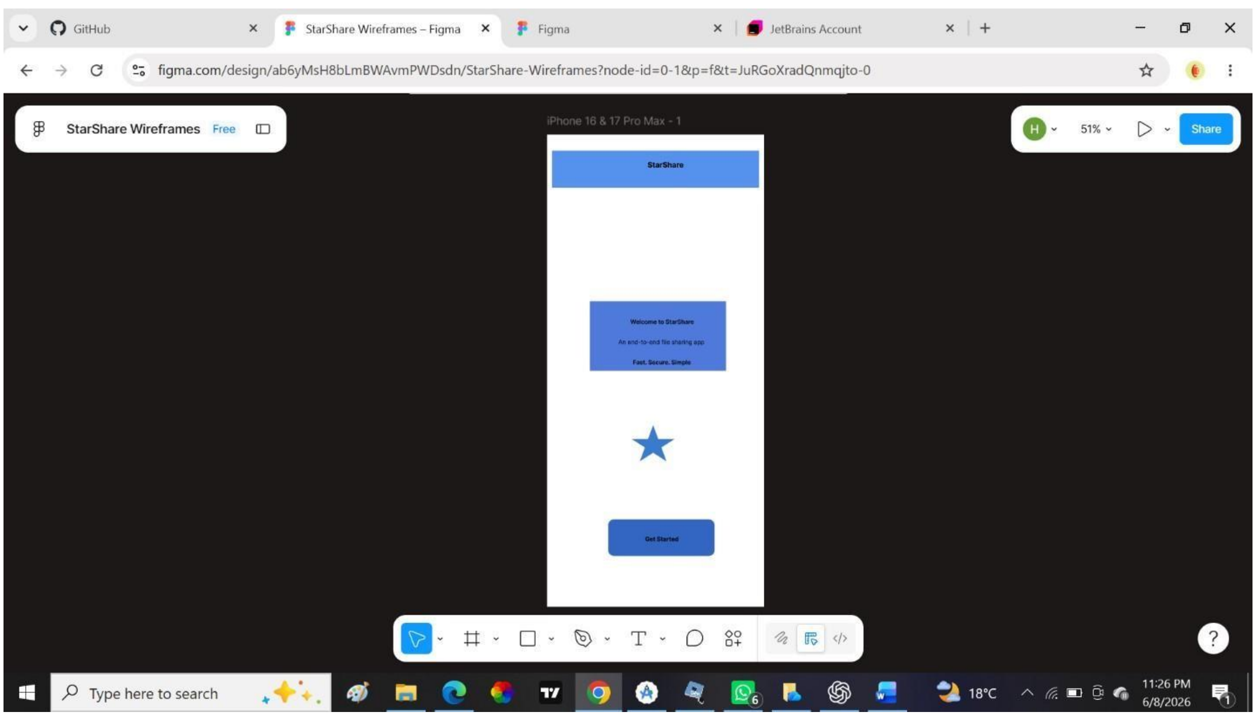
1. Android Studio
2. Flutter SDK
3. Android Emulator
4. Phone
5. Dart SDK
6. Figma

Personal Reflection

I Successfully installed and configured the development environment for the StarShare project. I gained practical experience setting up Flutter, Android Studio, and GitHub, which established a solid foundation for the development process.

Week 2: Mobile Development Environment **Practical Activities Undertaken**

1. Installed and configured Figma.
2. Explored mobile application UI/UX design principles.
3. Created project wireframe layouts in Figma.
4. Designed Splash Screen wireframe.
5. Designed Welcome/Onboarding Screen wireframe.
6. Designed Login Screen wireframe.
7. Designed Registration Screen wireframe.
8. Reviewed mobile application navigation structures.
9. **Figure 1:** Onboarding Screen Wireframe.



10. **Figure 2:** Splash Screen Wireframe.

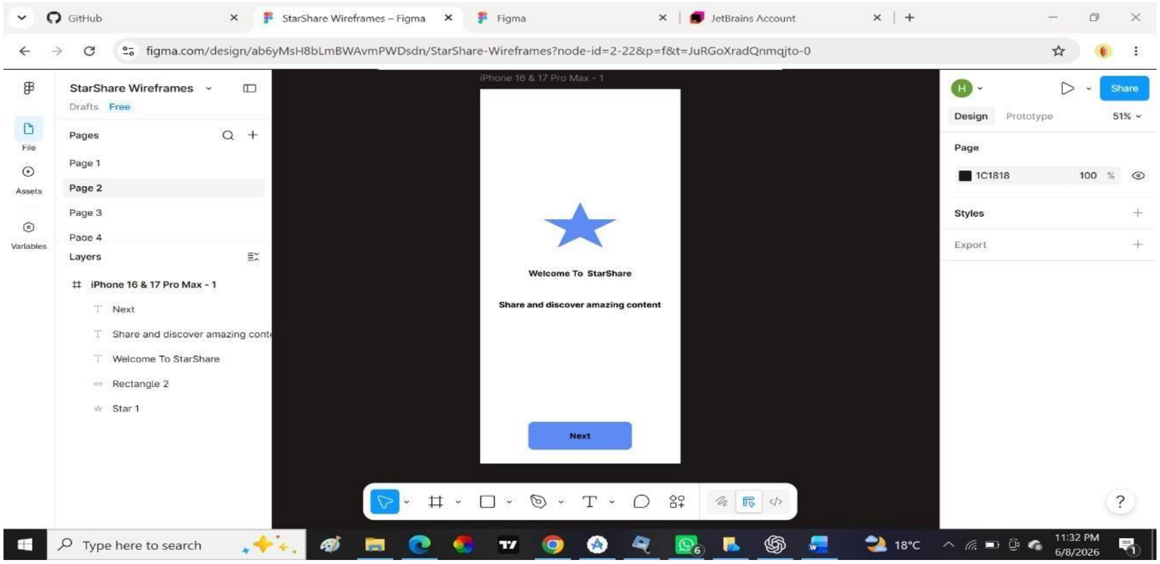


Figure 3: Login screen Wireframe

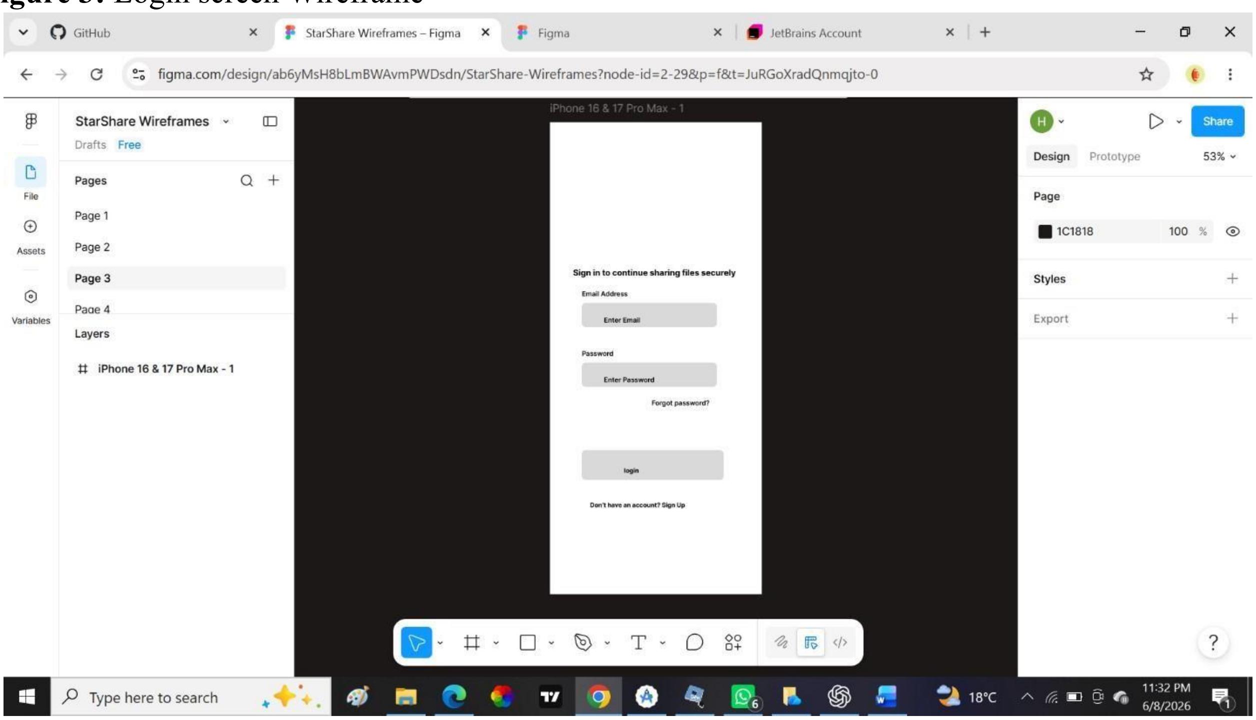
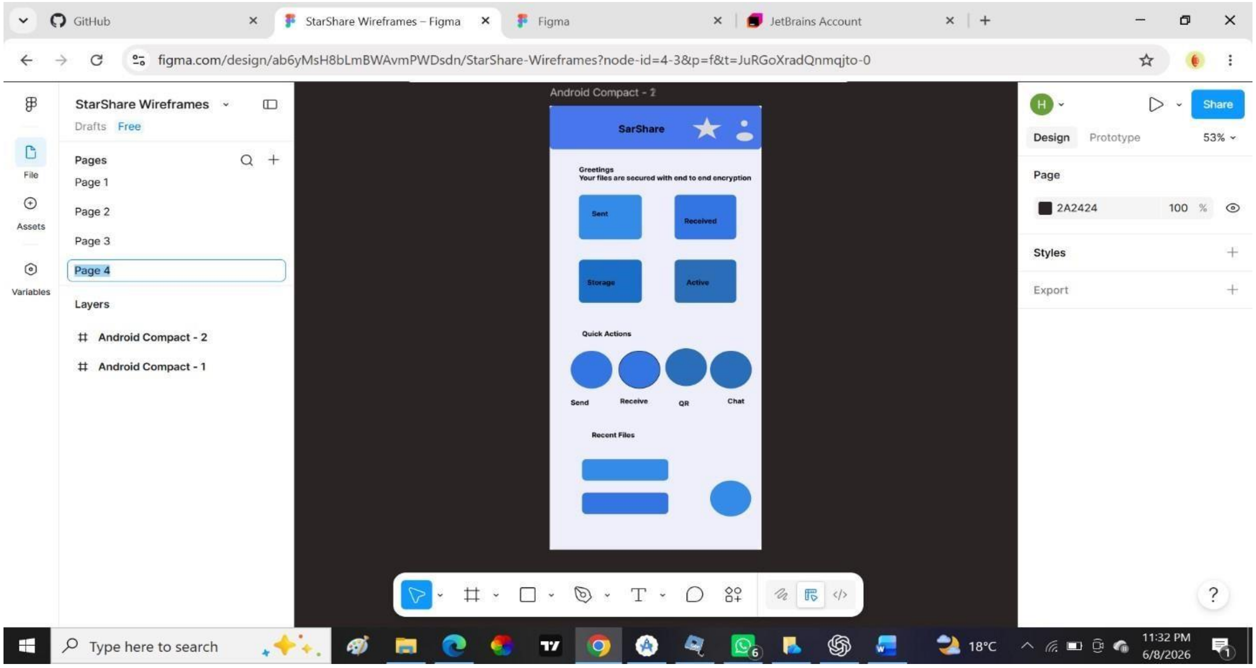


Figure 4: Home Screen Wireframe



Tools used.

1.Figma

2. Android studio (for references)

Personal Reflection

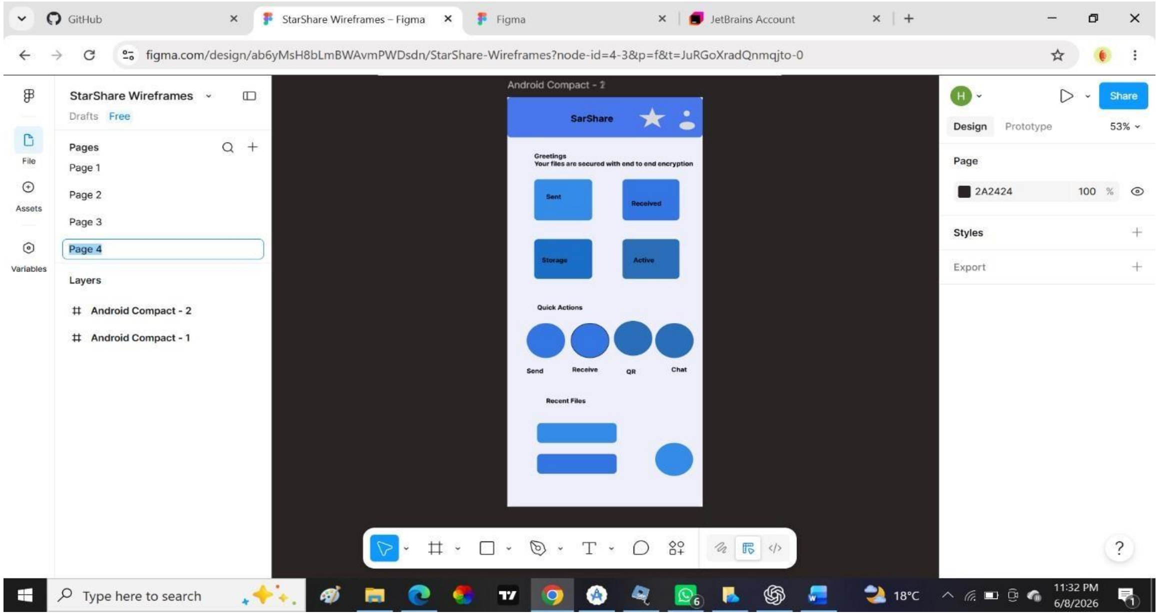
I developed wireframes using Figma and applied UI/UX design principles to improve usability. This activity provided a visual representation of the application structure and guided future development decisions.

Week 3: User Interface Design

Practical Activities Undertaken

1. Refined application wireframes.
2. Designed Home Dashboard interface.
3. Designed File Upload section
4. Designed Storage section
5. Improved navigation flow between screens.
6. Standardized buttons, icons, and layout structures.
7. Reviewed interface consistency and usability.

Figure 1: home screen



Tools/Software Used

1. Figma
2. Android Studio
3. Flutter SDK

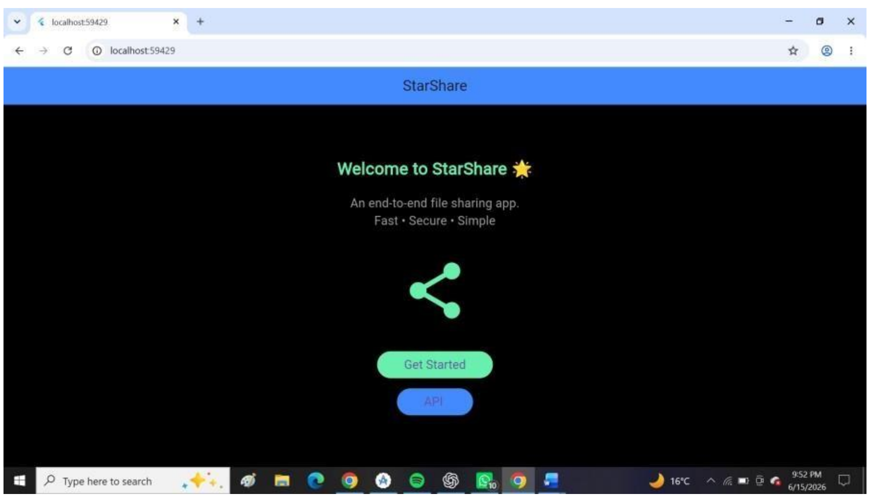
Personal Reflection

I refined the user interface design and improved consistency across all screens. The finalized wireframes provided a clear development roadmap and enhanced the overall usability of the StarShare application.

Week 4: Event Handling and User Interaction **Practical Activities Undertaken**

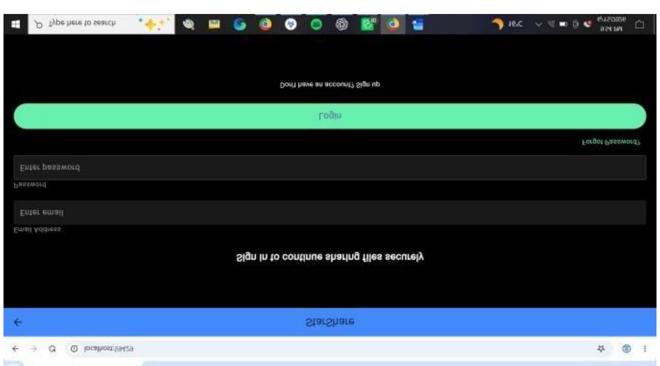
1. Implemented button click events in Flutter UI.
2. Added navigation between screens using routes.
3. Implemented form validation for login and registration.
4. Handled user inputs using TextEditingController.
5. Added loading indicators during user interactions.
6. Implemented basic error handling for invalid inputs.

Figure1



The get started button leads you to onboarding page.

Figure 2



The login button takes you to home page.

Tools used:

1. Flutter SDK
2. Android Studio
3. Dart Programming Language

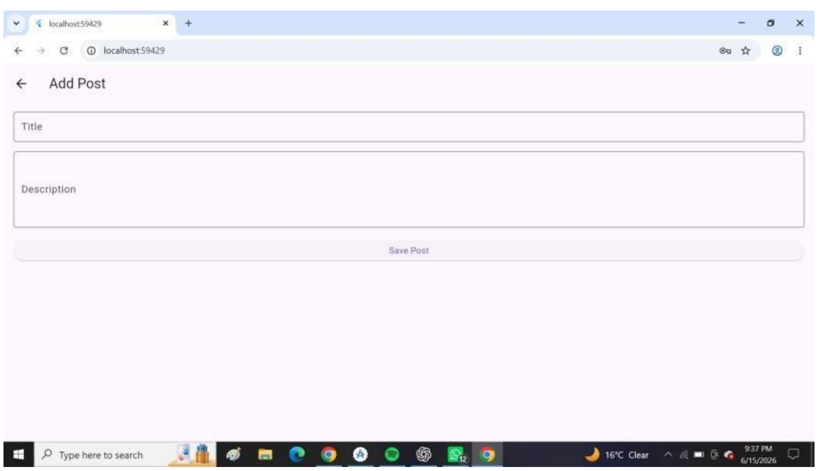
Personal Reflection

I learned how to handle user interactions in Flutter using event listeners, navigation, and form validation. This improved the responsiveness and usability of the application.

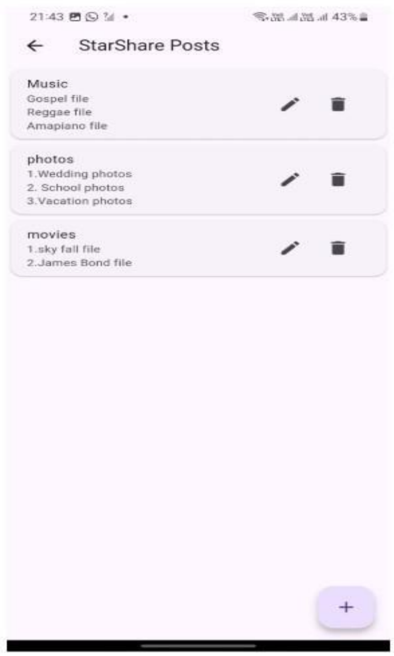
Week 5: Data Management

Practical Activities Undertaken

1. Created data models for users and posts.
2. Implemented local state management using setState().
3. Integrated API service for data fetching and posting.
4. Parsed JSON responses from backend.
5. Displayed dynamic data using ListView.builder.
6. Handled loading and error states in data rendering.



Allows adding and saving of posts



1 of 2

This one shows the editing and the deleting part of the system.

Tools Used

1. Flutter SDK
2. Dart
3. API Service (HTTP package)
4. Android Studio

Personal Reflection

I gained practical experience in managing and displaying dynamic data in Flutter applications. I also learned how to integrate backend APIs and handle JSON data effectively.

Week 6: Networking and APIs

Practical Activities Undertaken

1. Created REST API service for backend communication.
2. Implemented HTTP GET and POST requests in Flutter.

3. Connected Flutter application to backend server.
4. Handled API responses and errors.
5. Implemented asynchronous programming using async/await.
6. Tested network connectivity and data flow.

Figure1

This one shows a button for API for backend communication.

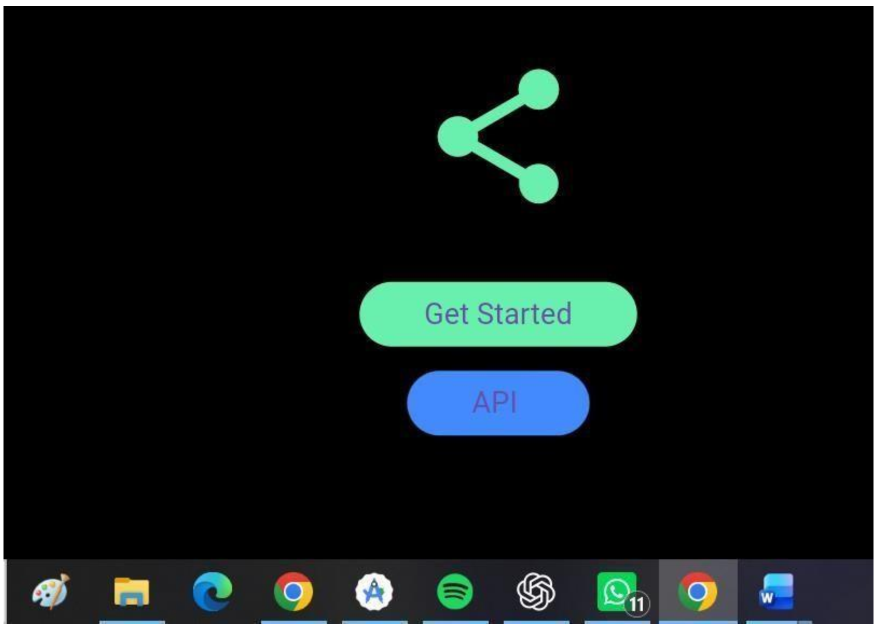


Figure 2
This shows when the internet access is denied, the server cannot be reached.

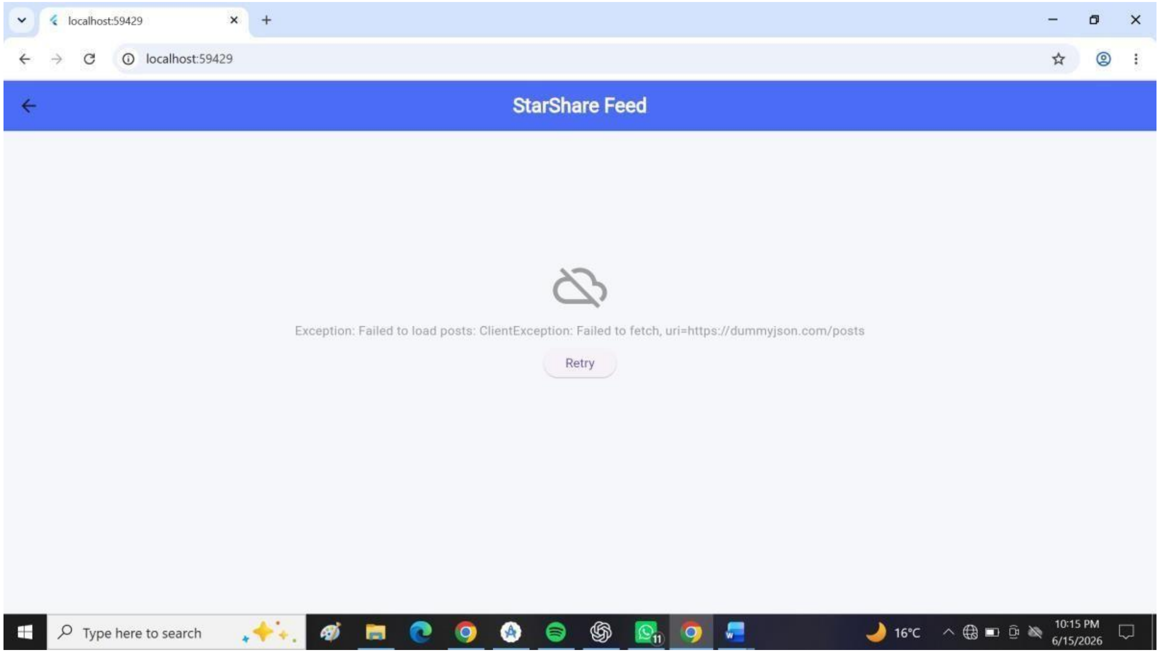
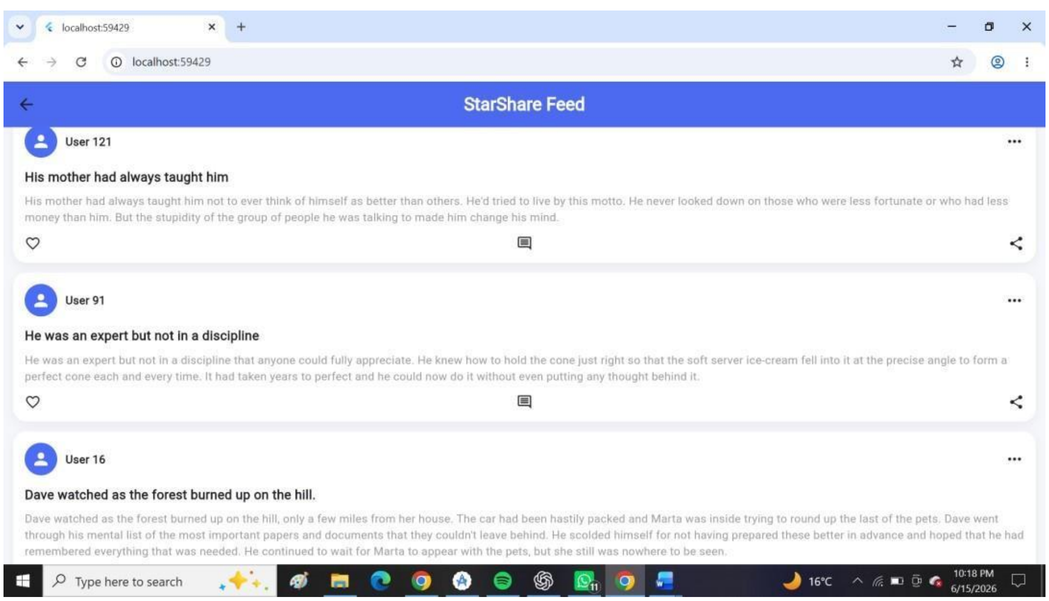


Figure 3

1. Handled API responses and the internet error and reached the information.



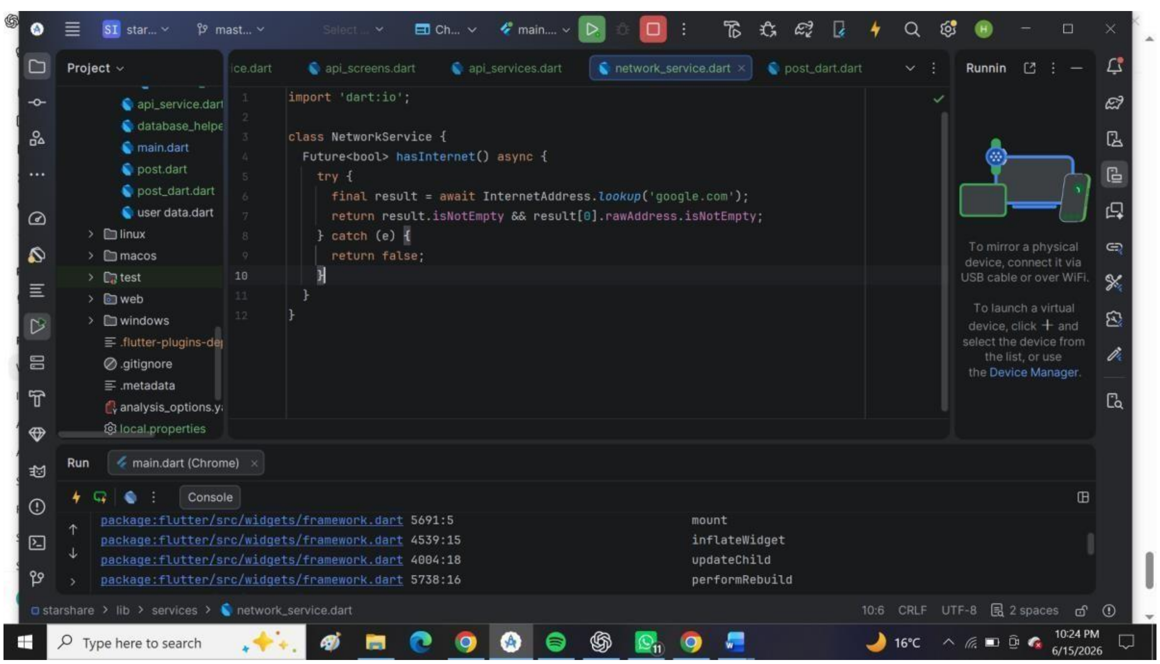
Tools

1. Flutter SDK
2. Dart
3. HTTP package
4. Backend API (Node.js / PHP / Firebase if used)
5. Android Studio

Personal Reflection

I learned how to connect a mobile application to external APIs and manage real-time data exchange. This improved my understanding of networking, asynchronous programming, and backend integration.

The network connection string code



Why does my system (Starshare) require networking?

StarShare requires networking to upload, sync, and retrieve files and user data between devices and the server.

Week 7: Application Architecture

Same as week 1

Application: StarShare File Sharing App

TASK 1: STORAGE METHOD COMPARISON (STARSHARE)

StarShare uses different storage methods depending on functionality.

Storage Method	Usage in StarShare	Advantages	Disadvantages
Shared Preferences	Stores user login status	Fast and simple	Not secure for sensitive data
SQLite Database	Stores file transfer history	Works offline and structured	Requires SQL knowledge
Firebase	Handles user authentication	Secure and real-time	Requires internet connection
Internal Storage	Stores received files	Private and offline access	Limited storage space

TASK 2: DATABASE DESIGN (STARSHARE SYSTEM)

System Overview

StarShare is a mobile application used for file sharing between devices using WiFi connection.

USERS TABLE (FIREBASE AUTH)

Field	Type	Description
user_id	STRING	Unique Firebase user ID

Field	Type	Description
email	STRING	User email address
password	STRING	Managed securely by Firebase Authentication

TRANSFERS TABLE (LOCAL SQLITE DATABASE)

Field	Type	Description
transfer_id	INTEGER	Primary key
file_name	TEXT	Name of file transferred
file_size	INTEGER	Size of file in bytes
direction	TEXT	SEND or RECEIVE
ip_address	TEXT	Device IP address
timestamp	TEXT	Date and time of transfer

SQL IMPLEMENTATION

```
CREATE TABLE Transfers (   transfer_id INTEGER
PRIMARY KEY AUTOINCREMENT,
  file_name TEXT,
file_size INTEGER,
direction TEXT,
ip_address TEXT,
timestamp TEXT
```

);

RELATIONSHIP DESIGN

Each user can perform multiple file transfers.

Relationship:

User (1) -------- (Many) Transfers

NORMALIZATION

First Normal Form (1NF):

- All fields contain atomic values

Second Normal Form (2NF):

- All attributes depend on the primary key

Third Normal Form (3NF):

- No transitive dependencies exist

PASSWORD SECURITY (FIREBASE)

StarShare uses Firebase Authentication.

- Passwords are NOT stored manually
- Firebase automatically encrypts and hashes passwords

Example:

Password: 123456
Stored: Encrypted hash value (Firebase managed)

TASK 3: MINI PROJECT (STARSHARE SYSTEM DESIGN)

Storage Methods Used

StarShare uses a hybrid storage system:

- Firebase Authentication
- SQLite Database
- Internal Storage

REASONS FOR SELECTION

Firebase:

- User authentication
- Secure login system
- Cloud-based storage support **SQLite:**
- Stores file transfer history
- Works offline

Internal Storage:

- Stores received files locally on device

DATABASE STRUCTURE

Users Table (Firebase)

- user_id
- email

Transfers Table (SQLite)

- transfer_id
- file_name
- file_size
- direction
- ip_address
- timestamp

EXPECTED OUTPUT

The StarShare application should:

- Allow user registration and login • Discover nearby devices
- Send files using WiFi connection
- Receive and store files locally

- Save transfer history
- Display sent and received file logs

CONCLUSION

StarShare demonstrates real-world mobile application networking using:

- Client-server architecture (TCP sockets)
- Device discovery (UDP broadcast)
- Cloud authentication (Firebase)
- Local data storage (SQLite)

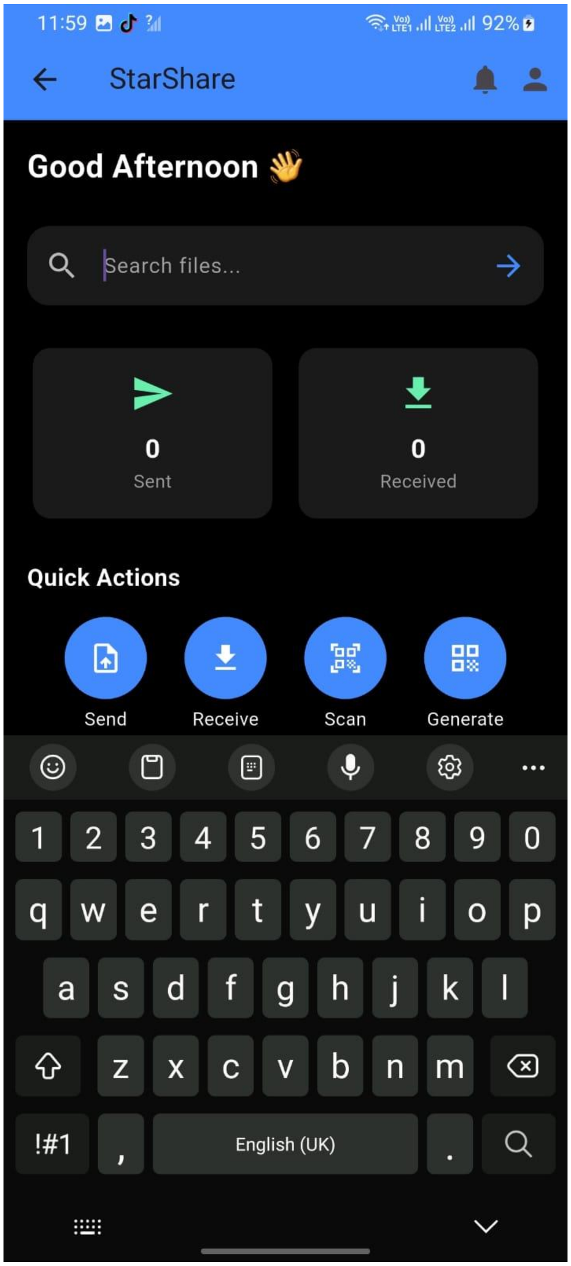
Week 8:

Task 1: Keyboard Input

Implemented:

- Search TextField
- Input validation using InputValidator
- SnackBar for empty search
- Keyboard hides after search

Figure 1



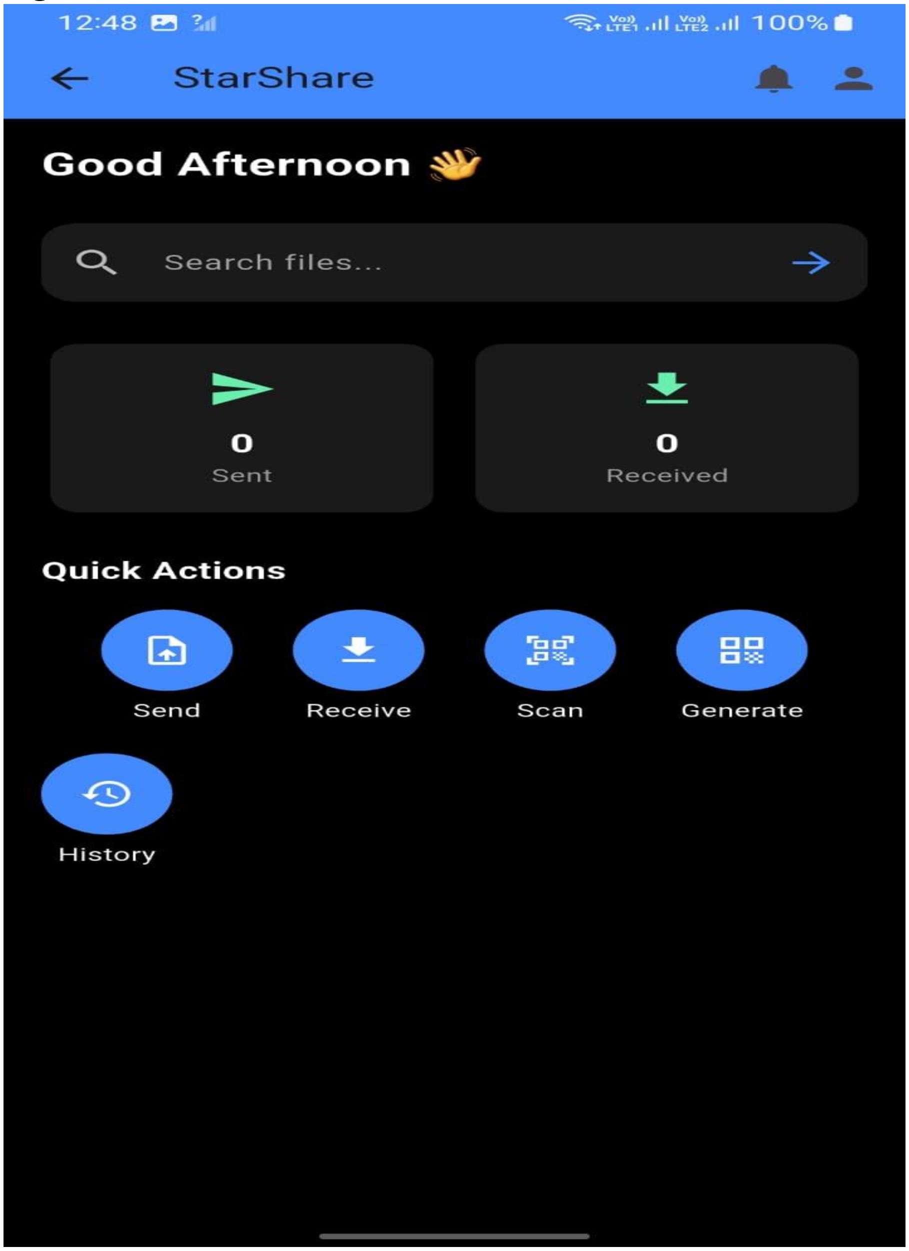
The screenshot above proves the keyboard inputs where a user can enter the name of the file and search it.

Task 2: Tap Events

Implemented separate methods for handling user actions:

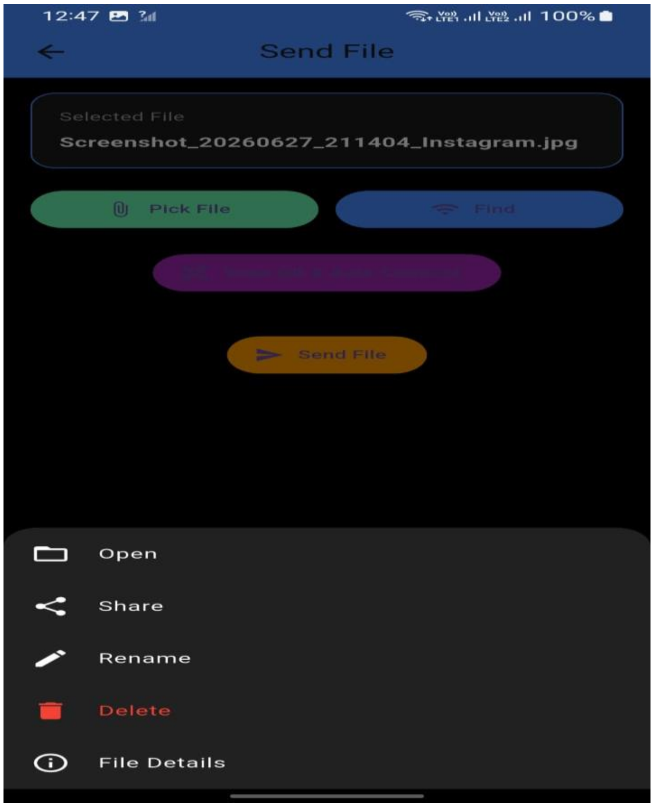
- Send
- Receive
- Scan/Connect
- Generate QR
- History

Figure 2



Task 3: Long Press
Write
Task 3: Long Press Gesture
Implemented long press on the Selected File card.
Options displayed:

- Open
- Share
- Rename
- Delete
- File Details

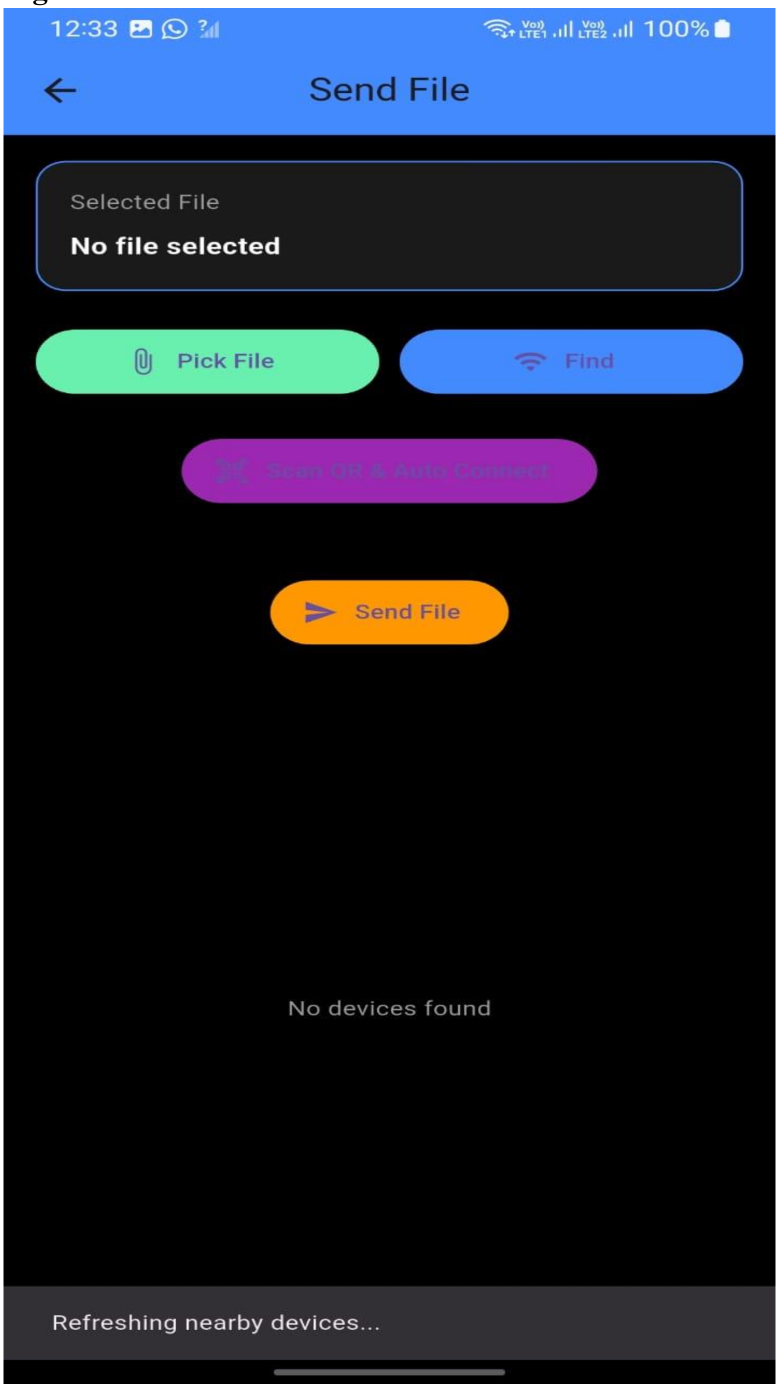
Implemented using the GestureHandler class.
Figure 3.


NB: When selected file is long pressed, the above options are provided at the bottom of the screen.

Task 4: Swipe Down Gesture
Implemented swipe down to refresh nearby devices.
Calls:

- discoverDevices()
- Displays SnackBar "Refreshing nearby devices..."

Figure 4.



When user swipes down, the list of present devices are refreshed.

Week 9: Device Features and Sensors

Same as week 1

Week 10: Notifications and Background Services

Same as week 1

Week 11: Mobile Security

Same as week 1

Week 12: Testing and Debugging

Same as week 1

Week 13: Application Deployment

Same as week 1

Week 14: Final Project Presentation

Same as week 1

Semester Project Tracking

Project Title

Project Objectives

Project Requirements

Development Milestones

Testing Results

Lessons Learned

Future Improvements

End of Semester Reflection

Summary of Skills Acquired

Most Interesting Topic Major Challenges Faced

How Challenges Were Overcome

Professional Growth Achieved

Overall Course Experience

Final Approval

Student Name: Haron Mutai

Student Signature:

Date: 9th June 2026

Lecturer Name: Michael Nyoro

Lecturer Remarks: ______________________

Date: ___________________________________